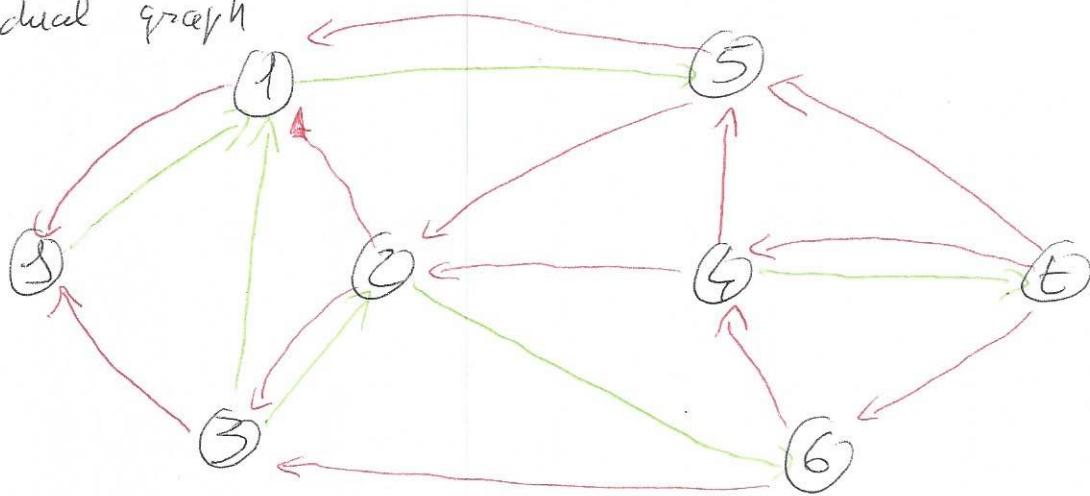


Residual graph



augmenting path with the min. number of arcs

$$s - 1 - 5 - 2 - 6 - 4 - t$$

$$\theta = \min \{ 8-5, 6-3, 4, 5-0, 2, 7-3 \} = 2$$

Updated flow

$$x_{s1} = 5+2$$

$$x_{15} = 3+2$$

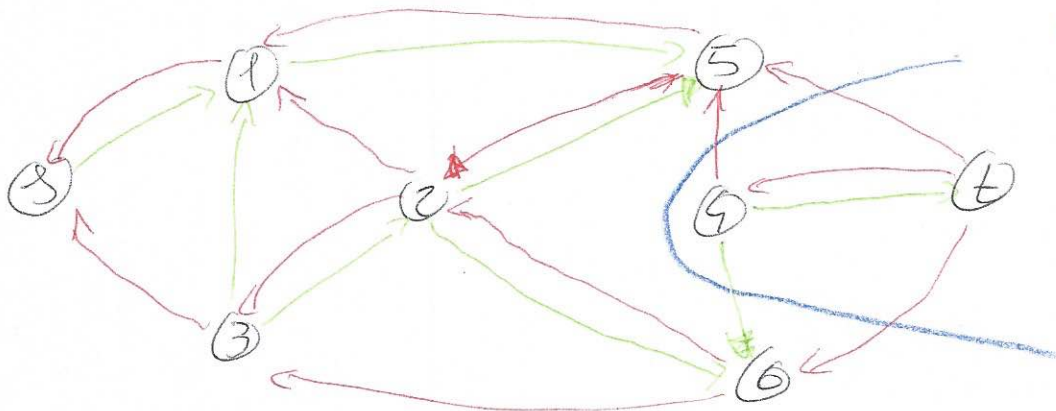
$$x_{25} = 4-2$$

$$x_{26} = 0+2$$

$$x_{46} = 2-2$$

$$x_{4t} = 4+2$$

Updated Residual Graph



Edmonds and Karp augmenting path algorithm
complexity $O(m^2 u)$

[EX 2]

$$\text{Min } 15y_1 + 6y_2$$

$$-5y_1 + y_2 - y_3 = -1$$

$$3y_1 + 3y_2 - y_3 = 2$$

$$y_1, y_2, y_3 \geq 0$$

$$y_1 (15 + 5x_1 - 3x_2) = 0$$

$$y_2 (6 - x_1 - 3x_2) = 0$$

$$y_3 (x_1 + x_2) = 0$$

$$x' = \begin{bmatrix} -3 \\ 3 \end{bmatrix} \quad I = \{2, 3\} \Rightarrow y_1 = 0$$

$$\begin{cases} y_2 - y_3 = -1 \\ 3y_2 - y_3 = 2 \end{cases}$$

$$y_2 = 3/2 \quad y_3 = 5/2$$

$$y' = [0, 3/2, 5/2] \geq 0 \text{ dual feasible}$$

x' does not satisfy constraint 1

$$-5(-3) + 3(3) \not\leq 15$$

x' and y' are not optimal

$$x'' = \begin{bmatrix} -3/2 \\ 5/2 \end{bmatrix} \quad I = \{1, 2\} \Rightarrow y_3 = 0$$

$$\begin{cases} -5y_1 + y_2 = -1 \\ 3y_1 + 3y_2 = 2 \end{cases}$$

$$y_1 = 5/18 \quad y_2 = 7/18$$

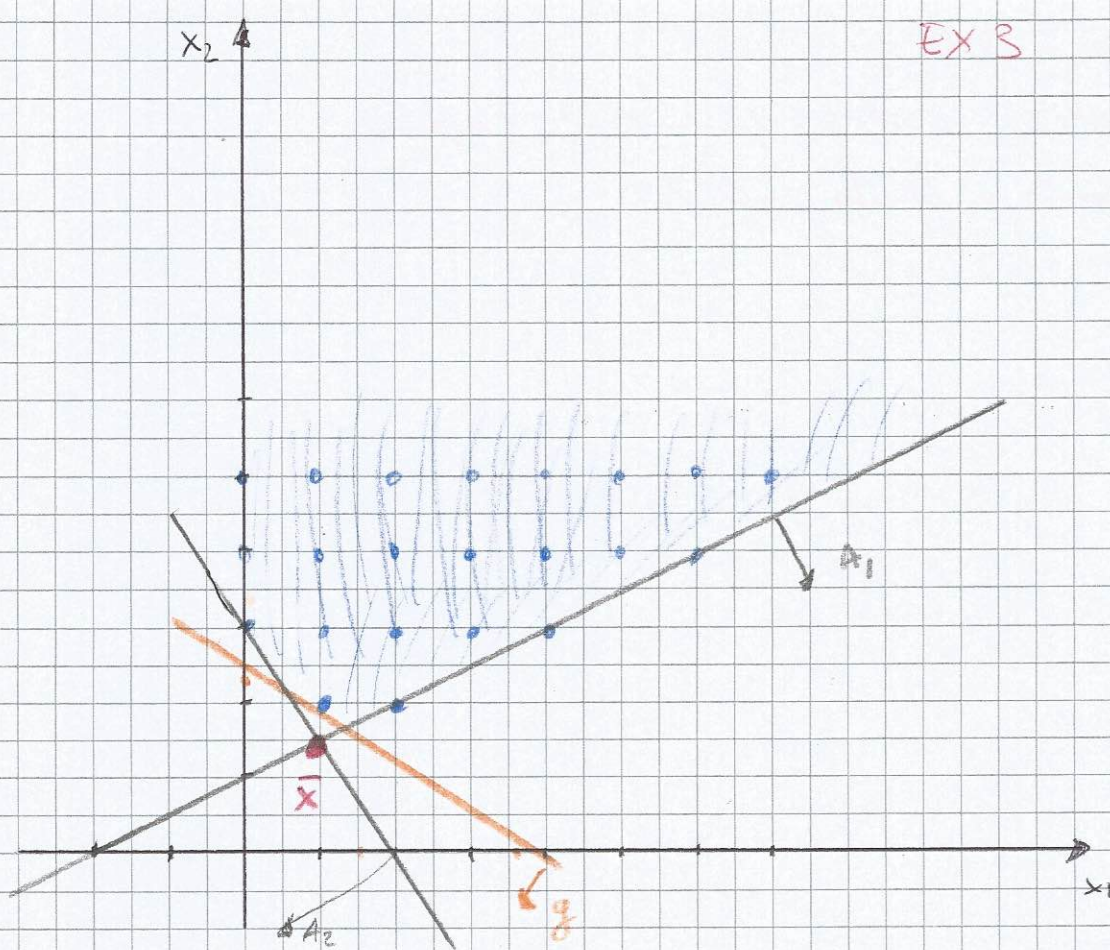
$$y'' = [5/18, 7/18, 0] \geq 0 \text{ dual feasible}$$

x'' satisfies constraint 3

$$-(-3/2) + (5/2) \leq 0$$

x'' and y'' are optimal

Ex 3



$$\max -x_1 - x_2$$

$$x_1 - 2x_2 + x_3 = -2$$

$$-3x_1 - 2x_2 + x_4 = -6$$

$$B = \{1, 2\} \quad A_B = \begin{bmatrix} 1 & -2 \\ -3 & -2 \end{bmatrix} \quad A_N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A_B^{-1} = \begin{bmatrix} 1/4 & -1/4 \\ -3/8 & -1/8 \end{bmatrix}$$

$$A_B^{-1}b = \bar{x} = \begin{bmatrix} 1 \\ 3/2 \end{bmatrix} = \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix} \rightarrow \text{fractional}$$

$$A_B^{-1} A_N = \begin{bmatrix} x_3 & x_4 \\ 1/4 & -1/4 \\ -3/8 & -1/8 \end{bmatrix}$$

$$\text{cut } g \cdot x \leq \gamma$$

$$x_2 + [-3/8]x_3 + [-1/8]x_4 \leq [3/2]$$

$$x_2 - x_3 - x_4 \leq 1$$

$$x_3 = -2 - x_1 + 2x_2$$

$$x_4 = -6 + 3x_1 + 2x_2$$

$$x_2 + 2 + x_1 - 2x_2 + 6 - 3x_1 - 2x_2 \leq 1$$

$$-2x_1 - 3x_2 \leq -7$$

(43 41)

ξ is a growing feasible direction for $\bar{x}$ if

$x' = \bar{x} + \lambda \xi$ is a new feasible point
with a better objective function
for a value $\lambda > 0$

1) if in $\bar{x}$ there are no active constraints
 $c\xi > 0$ is the growing condition
and the feasibility is always guaranteed

2) if $I = \{i: A_i \bar{x} = b_i\} \neq \emptyset$ then

$c\xi > 0$
 $A_i \xi \leq 0$ are the growing feasible direction
conditions

consider a new point $x' = \bar{x} + \lambda \xi$

$$c x' > c \bar{x}$$

$$c \bar{x} + \lambda c \xi > c \bar{x}$$

$$c \xi > 0$$

growing conditions

$$A_i x' \leq b_i$$

$$i \in I$$

$$A_i \bar{x} + \lambda A_i \xi \leq b_i$$

since $i \in I$ $b_i = A_i \bar{x}$

$$b_i + \lambda A_i \xi \leq b_i$$

$$A_i \xi \leq 0$$

$$i \in I$$

Ex 5

set $I := 1..10$;

set $I_bar := \{ I \text{ union } \{0\} \}$;

var $x \{ I_bar \} z = 0$;

param $b \{ I, I_bar \}$;

s.t. c1:

$\text{sum}\{i \text{ in } I\} (\text{sum}\{j \text{ in } I_bar\} (b[i,j] * x[j])) \leq$

$\text{sum}\{i \text{ in } I\} (b[i,0] * x[0]);$

t_i = starting time of activity $i \in A$ Ex 6

$$\min t_n - t_1$$

$$t_i + p_i \leq t_j \quad \forall (i,j) \in P$$

Variant[†]

s_i = saved days for activity i

$$\min \sum_{i \in A} c_i s_i$$

$$t_n - t_1 \leq D$$

$$t_i + (p_i - s_i) \leq t_j \quad \forall (i,j) \in P$$

$$s_i \leq p_i - p'_i \quad \forall i \in A$$

$$s_i \geq 0$$

$$N = \{0, F, 1..u\}$$

$$A = \{(i, j) : i, j \in \{1..u\}, b_i < a_j\} \cup \quad *$$

$$\{(0, i) : i \in \{1..u\}\} \cup \{(j, F) : j \in \{1..u\}\}$$

cost of the arcs:

$$c_{0, i} = 0 \quad i = 1..u$$

$$c_{i, j} = r_i \quad i = 1..u \quad j = 1..u, F$$

Any path from 0 to F correspond to a feasible choice of fields to visit

The maximum cost path corresponds to the highest revenue choice

The graph is acyclic due to *

The problem can be solved in linear time with SP/ Acyclic